

Econometrics I

Lecture 11: Count Data

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- ▶ Dependent variable measures a count $y = 0, 1, 2, \dots$
- ▶ Examples: number of units purchased, number of doctor visits, number of children (fertility).

Poisson distribution

- ▶ **Poisson process**: events happen independently over time at rate λ per unit time. The **Poisson distribution** gives the distribution over the number of events that happen.

- ▶ Probability mass:

$$Pr(Y = y \mid \lambda) = \frac{e^{-\lambda} \lambda^y}{y!}$$

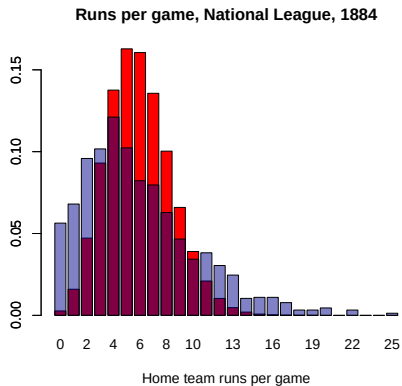
The Poisson distribution just has one parameter:

$$\mathbb{E}[y] = \lambda$$

$$\mathbb{V}[y] = \lambda$$

- ▶ The Poisson distribution features **equidispersion**, mean=variance.
- ▶ It's more common to see **overdispersion** in count data (variance > mean) (which sometimes leads to *negative binomial* models).

Poisson distribution



Plotted Poisson distribution (red) is for λ equal to the average runs per game.

Empirical probability mass function is in blue.

Poisson regression (MLE)

- Let the Poisson rate vary with observable characteristics $\mathbf{x}$:

$$\lambda(\mathbf{x}) = \mathbb{E}[y \mid \mathbf{x}] = \exp(\mathbf{x}'\boldsymbol{\beta}).$$

Note that this ensures that $\lambda(\mathbf{x}) > 0$.

- Log likelihood follows from Poisson pdf:

$$\ell(y, x) = \frac{e^{-\lambda} \lambda^y}{y!}$$

$$\ln \ell(y, \mathbf{x}) = -\exp(\mathbf{x}'\boldsymbol{\beta}) + y\mathbf{x}'\boldsymbol{\beta} - \ln y!$$

$$\ln L(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}) = \sum_{i=1}^n \{-\exp(\mathbf{x}_i'\boldsymbol{\beta}) + y_i\mathbf{x}_i'\boldsymbol{\beta} + \ln y_i!\}$$

- Log likelihood:

$$\ln L(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}) = \sum_{i=1}^n \{-\exp(\mathbf{x}_i' \boldsymbol{\beta}) + y_i \mathbf{x}_i' \boldsymbol{\beta} + \ln y_i!\}$$

- First-order condition for maximum:

$$\sum_{i=1}^n \{-\exp(\mathbf{x}_i' \boldsymbol{\beta}) \mathbf{x}_i + y_i \mathbf{x}_i\} = 0$$

$$\sum_{i=1}^n \{(y_i - \exp(\mathbf{x}_i' \boldsymbol{\beta})) \mathbf{x}_i\} = 0$$

- Log likelihood function is globally concave, so computer will generally converge to solution quickly.

- Marginal effects are given by

$$\frac{\partial \mathbb{E}[y \mid \mathbf{x}]}{\partial x_j} = \exp(\mathbf{x}'\boldsymbol{\beta}) \beta_j = \mathbb{E}[y \mid \mathbf{x}] \beta_j$$

- Alternatively,

$$\frac{1}{\mathbb{E}[y \mid \mathbf{x}]} \frac{\partial \mathbb{E}[y \mid \mathbf{x}]}{\partial x_j} = \beta_j,$$

which makes it clear that coefficients can be interpreted as proportional change in mean for one unit change in covariate x_j .

- First-order condition for maximum:

$$\sum_{i=1}^n \{(y_i - \exp(\mathbf{x}_i' \boldsymbol{\beta})) \mathbf{x}_i\} = 0$$

- Note this moment condition is satisfied if $\mathbb{E}[y_i \mid \mathbf{x}_i] = \exp(\mathbf{x}_i' \boldsymbol{\beta})$.
- MLE estimation consistent even if $\mathbb{V}[y_i \mid \mathbf{x}_i] \neq \exp(\mathbf{x}_i' \boldsymbol{\beta})$.
- $\Rightarrow$ we might want to estimate Poisson MLE even if we don't believe in equidispersion.

- ▶ If we don't want to impose equidispersion, then we shouldn't rely on MLE standard errors; we should instead look at this as GMM estimation, and compute standard errors using standard sandwich formula (more on this next semester).
- ▶ With equidispersion (full Poisson model), see MLE slides for standard errors formulas.
- ▶ There can be a big difference between the two!

- Instead of imposing $\mathbb{V}[y \mid \mathbf{x}] = \mathbb{E}[y \mid \mathbf{x}]$, we can let

$$\mathbb{V}[y \mid \mathbf{x}] = \gamma \mathbb{E}[y \mid \mathbf{x}] = \gamma \exp(\mathbf{x}'_i \boldsymbol{\beta}),$$

and estimate

$$\hat{\gamma} = \frac{1}{n - k} \sum_i \frac{(y_i - \exp(\mathbf{x}'_i \boldsymbol{\beta}))^2}{\exp(\mathbf{x}'_i \boldsymbol{\beta})}.$$

Note that this is just a standard estimator of the variance, divided by our formula for the mean.

- This makes sense for *estimation*, and γ can be used in sandwich estimator for consistent standard errors. But what about for *simulation* – what is the distribution of the data generating process with $\gamma \neq 1$?

- Let

$$y \sim \text{Poisson}[\lambda\nu]$$

$$\nu \sim \text{Gamma}[\mu = 1, \sigma^2 = \alpha].$$

Then we say that y has a negative binomial distribution with mean λ and variance $\lambda + \alpha\lambda^2$.

- The negative binomial distribution is typically described as the number of failures observed before we see r successes when doing independent Bernoulli trials with success rate p . The above has the same distribution as $r = \frac{1}{\alpha}$ and $p = \frac{\lambda\alpha}{1+\lambda\alpha}$.
- Note that, with either of the above descriptions, it's easy to describe and simulate the full distribution or data generating process (you can look up the distribution functions on Wikipedia).

Negative binomial distribution: estimation

- ▶ When we're worried about overdispersion, it makes sense to assume a negative binomial distribution instead of a simple Poisson.
- ▶ We could use MLE for the negative binomial distribution. This is the most efficient thing if the specification is correct (MLE is always asymptotically efficient for correctly specified models).
- ▶ However, we could still do Poisson regression (noting that the negative negative binomial extension does not change the mean, and Poisson regression is consistent as long as mean is correctly specified). In this, can estimate model of (conditional) mean using Poisson regression.
- ▶ If you're only interested in how the mean of y depends on x , you're done after estimating a Poisson regression. If you want to do simulations or compute standard errors on the variance parameter (robust to overdispersion), you should estimate the negative binomial model using a GMM or MLE estimator.

- ▶ Poisson is nested within negative binomial with $\alpha = 0$. Thus, likelihood ratio test applies and effectively serves as a test for overdispersion.
- ▶ As Lancaster (1997) points out, Poisson model with fixed effects has incidental parameters, but they are not problematic.
 - Recall in binary choice logit model, we can handle fixed effects by constructing conditional likelihood function. For Poisson models, we can just use the regular likelihood function with fixed effects, and those fixed effects don't create a problem for our standard errors. The Poisson model and linear regression model have this in common.
- ▶ Poisson regression can be used for things that don't have the Poisson distribution's equidispersion, but you shouldn't use the MLE estimator's standard errors in that case.

Text Analysis

- ▶ [Gentzkow, Kelly, and Taddy \(2019\)](#) provides a good overview from a social science perspective.
- ▶ Text analysis generally built on either **multinomial choice models** (there is a probability distribution over what each word is) or **count data models** (there is a probability distribution over counts of words or phrases).
- ▶ It's common to model different distributions for different topics. For example, in the latent Dirichlet allocation,
 - Documents are drawn from topics. A probability distribution over topics is estimated. We don't pre-specify what the topics are (**unsupervised learning**), though we might pre-specify the number of topics.
 - Each topic is associated with a probability distribution over words (which we estimate).
 - Post-estimation, the topics can typically be interpreted based on which words have relatively high frequencies. Application: grouping papers or patent applications by field.

Gentzkow, Shapiro, and Taddy (2019)

- ▶ Measuring divergence in speech patterns of US political parties. “Speaking different languages.”
- ▶ Documents increasing partisanship since 1994 (Newt Gingrich). A rare example of great descriptive econometrics.
- ▶ Challenges:
 - There are lots of words, many of them used very rarely.
 - Standard multinomial logit model computationally intractible.

- ▶ $\mathbf{x}_{i,t}$ is a vector of speaker i 's characteristics in session t .
 - state, chamber, gender
- ▶ $\mathbf{q}_t^P(\mathbf{x}_{i,t})$ is a J -dimensional (very large) vector of probabilities of using phrases. $P \in \{R, D\}$ denotes party.
- ▶ We're interested in whether $\mathbf{q}_t^D$ differs from $\mathbf{q}_t^R$, and whether the difference between the two increases over time.

- Define

$$\rho_{jt}(\mathbf{x}) = \frac{q_{jt}^R(\mathbf{x})}{q_{jt}^R(\mathbf{x}) + q_{jt}^D(\mathbf{x})}$$

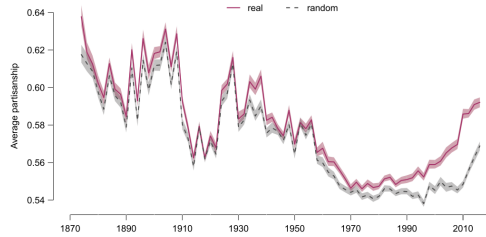
$$\pi_t(\mathbf{x}) = \frac{1}{2} \mathbf{q}_t^R(\mathbf{x}) \cdot \rho_t(\mathbf{x}) + \frac{1}{2} \mathbf{q}_t^D(\mathbf{x}) \cdot (1 - \rho_t(\mathbf{x}))$$

- $\rho_{jt}(\mathbf{x})$ is poserior probability that phrase j at time t comes from Republican.
- $\pi_t(\mathbf{x})$ aggregates these probabilities a natural measure of the partisanship. It's the probability you can guess the speaker's party correctly if they have characteristics x . $\bar{\pi}_t$ then averages over x .

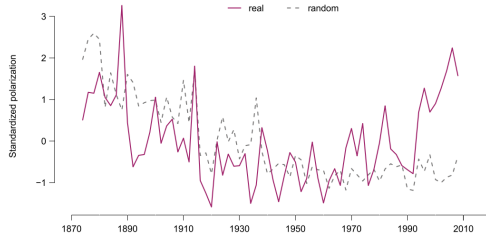
An important theme of our paper is that this and related estimators can be severely biased in finite samples even if $\mathbf{x}_{it} := \mathbf{x}_t$. Intuitively, partisanship will be high when the dispersion of the posteriors ρ_{jt} is large—that is, when some phrases are spoken far more by Republicans and others are spoken far more by Democrats. The MLE estimates the ρ_{jt} using their sample analogues $\hat{\rho}_{jt}$. However, sampling error will tend to increase the dispersion of the $\hat{\rho}_{jt}$ relative to the dispersion of the true ρ_{jt} . When the number of phrases is large relative to the volume of speech observed, many phrases will be spoken only a handful of times, and so may be spoken mainly by Republicans ($\hat{\rho}_{jt} \approx 1$) or mainly by Democrats ($\hat{\rho}_{jt} \approx 0$) by chance even if the true choice probabilities do not differ by party.

1. **Leave-out estimator:** Estimate $\mathbf{q}_t$ and ρ_t using different samples.
2. **Penalized estimator:** parameterized choice model. Like Lasso, incorporate penalty for each non-zero parameter.
 - Instead of multinomial logit estimator of frequencies, use Poisson model of counts. Note that Poisson mean $\exp(\mathbf{x}'\boldsymbol{\beta})$ is also the numerator of a logit model. Apparently avoiding the need to compute the denominator is a big deal computationally.

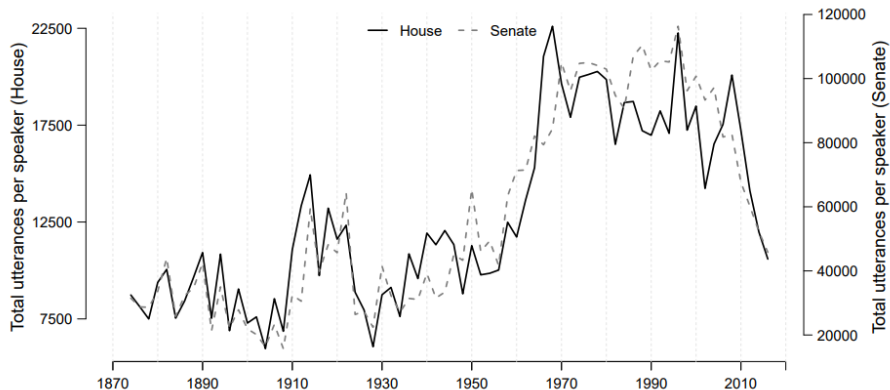
Panel A: Partisanship from Maximum Likelihood Estimator ($\hat{\pi}_t^{MLE}$)



Panel B: Polarization from Jensen et al. (2012)

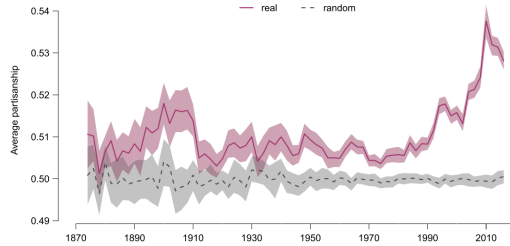


Online Appendix Figure 2: Average Verbosity per Speaker-Session



Notes: Plot shows, separately for each chamber and session, the average number of phrase (bigram) utterances across speakers in our estimation sample before making any vocabulary restrictions.

Panel A: Partisanship from Leave-out Estimator ($\hat{\pi}_t^{LO}$)



Panel B: Partisanship from Preferred Penalized Estimator ($\hat{\pi}_t^*$)

